

Cross-cohort candidate biomarkers in under-served diseases: a transparent multi-disease negative-results study

MEGA-PROGRAM-27 working analysis

25 September 2026 - working manuscript; not yet complete

Abstract

We evaluate expression-derived gene candidates across ten disease labels (nine with analyzable cohorts), including postpartum depression. A prespecified duplicate-sample audit materially changed earlier apparent replications. In the deduplicated analysis, top-50 signatures passed an empirical-null sign test in polycystic ovary syndrome (PCOS) and interstitial cystitis, though the latter has a small validation cohort; five other testable disease signatures did not pass. A named preeclampsia candidate, GNG2, did not replicate. Independent fresh RNA-seq cohorts falsified the directional ST3GAL2 prediction. A held-out cohort convolutional neural network (CNN) scored below logistic regression on average. The graph neural network (GNN) benchmark is within-project only; no published state-of-the-art comparison or validated new biomarker has yet been established. This manuscript is a working report, not a clinical test claim.

Contents

1	Scope and reproducibility	3
2	Data acquisition and splits	3
2.1	Complete retained split and screening exclusions	3
3	Effect estimates and derivation	5
4	Graph, CNN and evaluation design	6
5	Deduplicated results	6
6	Prospective prediction attempts and failures	8
6.1	Preeclampsia effect-estimator sensitivity	8
6.2	All deduplicated signature replications	8
7	Novelty and literature checks	9
7.1	Physical-network context for the fixed panels	9
8	Limitations and next work	10
9	Full cohort-level results and endpoint boundaries	10
9.1	Discovery availability and fragility	10
9.2	Full fresh-cohort disposition	10
9.3	All graph-ranking comparisons	11
9.4	Paired graph-baseline comparisons within the project	12
9.5	All held-out patient-classifier cohorts	13
9.6	Cohort-level uncertainty for the CNN comparison	13
9.7	PCOS follow-up with a null matched to selection	14

10 A second matched-tissue PCOS null result	14
11 Interstitial cystitis: a registered cross-tissue stress test	15
12 Position against published work	16
13 Published postpartum-depression genes against existing cohorts	17
14 Threats to interpretation and reproducibility contract	17
14.1 Independent study versus independent record	17
14.2 Protocol, tissue and phenotype	17
14.3 Multiplicity and model-selection boundaries	17
14.4 Runnable artifacts and audit trail	17
15 Limitations of the fixed sign-panel test	18
16 Verified primary-literature references	18
A Accession and tool ledgers	19
B Additional registered null tests	20

1 Scope and reproducibility

The repository contains a runnable Python command-line tool, hermetic tests, result tables, accession manifest, and the source for these figures. The project-level completion gates are not all met: 36 of 40 genuinely used scientific tools, 132 of 120 distinct fetched-and-used accession records, eleven displayed numbered formulas, no externally comparable benchmark break or defensible new named discovery, and fewer than the required 20 substantive manuscript pages. The 132-record tally follows the program’s record-level rule; it includes 32 GSM sample accessions nested in eight studies, so it is not 130 independent cohorts. The manifest’s unit is a unique accession-backed record, not a matrix of conditions counted several times. Counts are computed from the committed manifests. Long COVID lacked a usable human case/control expression series under the current rules; this is an absence of usable data in this search, not proof none exists.

2 Data acquisition and splits

GEO array discovery uses NCBI E-utilities search and summary, GEO series-matrix FTP, and platform annotations. A separate BioStudies entry E-GEOD-4707 describes a 14-sample placental array design with early/late-onset groups and pooled normal reference; it is a study-context cross-check, not an additional held-out validation cohort (<https://www.ebi.ac.uk/biostudies/api/v1/studies/E-GEOD-4707>). RNA-seq matrices from GEO supplement files are processed separately. Cases and controls are assigned only from explicit sample metadata, not from accession titles alone. We retain the list of failed series and label audits. Samples appearing under identical GSM identifiers in GEO subseries or superseries are removed by a results-blind deterministic rule. In a pair with overlapping labeled GSMs, the smaller series is dropped; ties drop the higher accession. The final split is in `results/split_final.csv`. This correction invalidates earlier apparent preeclampsia and leishmaniasis replication passes.

2.1 Complete retained split and screening exclusions

The retained series are not evenly distributed. The next table reports every discovery and validation accession surviving the sample-identity deduplication; the PPD validation addendum is a separate row in its own locked file and is not accidentally treated as new P6 evidence. A split is a study-level assignment, not a guarantee that its samples generalize to another setting. This explicit list also allows a reader to find any series-level omission without reverse-engineering the figures. The excluded-series table records the confounds that defeated case/control interpretation even when an assay matrix existed.

Table 1: Deduplicated discovery and validation allocation. This is the full retained-series split, not an extra sample count. The distinct PPD GSE290313 validation addendum and fresh RNA-seq tests are described separately.

Disease	GEO	Split	Cases	Controls
chagas	GSE84796	discovery	10	7
endometriosis	GSE23339	discovery	10	9
endometriosis	GSE25628	discovery	8	14
endometriosis	GSE35287	discovery	28	40
endometriosis	GSE47360	discovery	6	3
endometriosis	GSE51981	discovery	77	71
endometriosis	GSE6364	discovery	21	16
endometriosis	GSE7846	discovery	5	5
endometriosis	GSE120103	validation	18	18
endometriosis	GSE248593	validation	6	6
endometriosis	GSE58178	validation	6	6
endometriosis	GSE73622	validation	22	28
fibromyalgia	GSE67311	discovery	67	75

Disease	GEO	Split	Cases	Controls
interstitial_cystitis	GSE11783	discovery	10	6
interstitial_cystitis	GSE621	discovery	6	8
interstitial_cystitis	GSE57560	validation	4	3
leishmaniasis	GSE55664	discovery	25	10
leishmaniasis	GSE80008	discovery	18	12
leishmaniasis	GSE125993	validation	29	16
me_cfs	GSE14577-GPL96	discovery	8	7
me_cfs	GSE14577-GPL97	discovery	8	7
me_cfs	GSE16059	validation	32	44
pcos	GSE34526	discovery	7	3
pcos	GSE43322	discovery	24	7
pcos	GSE5090	discovery	9	8
pcos	GSE54248	discovery	4	4
pcos	GSE54250-GPL8979	discovery	4	4
pcos	GSE6798	discovery	16	13
pcos	GSE106724	validation	4	4
pcos	GSE124226	validation	4	4
pcos	GSE137684	validation	8	4
pcos	GSE80432	validation	4	8
postpartum_depression	GSE45603	discovery	16	32
preeclampsia	GSE10588	discovery	17	26
preeclampsia	GSE24129	discovery	8	8
preeclampsia	GSE25906	discovery	23	37
preeclampsia	GSE30186	discovery	6	6
preeclampsia	GSE35574	discovery	19	40
preeclampsia	GSE43942	discovery	5	7
preeclampsia	GSE48424	discovery	18	18
preeclampsia	GSE54400	discovery	35	9
preeclampsia	GSE54618	discovery	12	12
preeclampsia	GSE60438-GPL10558	discovery	35	42
preeclampsia	GSE60438-GPL6884	discovery	25	23
preeclampsia	GSE66273	discovery	12	5
preeclampsia	GSE73374	discovery	19	17
preeclampsia	GSE75010	discovery	22	135
preeclampsia	GSE91189	discovery	4	4
preeclampsia	GSE93839	discovery	11	12
preeclampsia	GSE147776	validation	13	8
preeclampsia	GSE149440	validation	66	314
preeclampsia	GSE186819	validation	12	13
preeclampsia	GSE190639	validation	6	13
preeclampsia	GSE94643	validation	8	8

Table 2: All series excluded during case/control phenotype screening. The matrix being available does not make a matched patient case/control comparison.

Disease	GEO	Exclusion reason
preeclampsia	GSE86200	vitamin D intervention vs control arm; not a preeclampsia contrast
preeclampsia	GSE25861	IUGR vs preterm-labour endothelial cells; no preeclampsia contrast

Disease	GEO	Exclusion reason
preeclampsia	GSE94644-GPL13497	case/control confounded with in-vitro decidualisation state
preeclampsia	GSE91077	case/control confounded with in-vitro decidualisation state
leishmaniasis	GSE77528	cases are post-treatment patients (months after anti-mony)
leishmaniasis	GSE43661	in-vitro macrophage infection model; not patient tissue
chagas	GSE128270	in-vitro HUVEC infection confounded with simvastatin pre-incubation
chagas	GSE13791	in-vitro smooth-muscle cell infection model; not patient tissue
pcos	GSE48301	case stromal vs control epithelial cells (cell-type confound)

3 Effect estimates and derivation

For cohort c and gene g , with n_1 case samples and n_0 controls, estimate the pooled sample variance as

$$s_{p,gc}^2 = \frac{(n_1 - 1)s_{1,gc}^2 + (n_0 - 1)s_{0,gc}^2}{n_1 + n_0 - 2}. \quad (1)$$

The standardized mean difference is $d_{gc} = (\bar{x}_{1,gc} - \bar{x}_{0,gc})/s_{p,gc}$. We correct its small-sample bias with $J(n) = 1 - 3/(4n - 9)$:

$$g_{gc} = J(n_1 + n_0)d_{gc}. \quad (2)$$

The plug-in within-study variance is

$$v_{gc} = \frac{n_1 + n_0}{n_1 n_0} + \frac{g_{gc}^2}{2(n_1 + n_0)}. \quad (3)$$

These formulas follow by first pooling the two unbiased sample variance estimates using their residual degrees of freedom, then standardizing the case-control mean contrast, then multiplying by the first-order Hedges correction. For k studies of a gene, define inverse-variance fixed weights $w_c = v_c^{-1}$ and pooled fixed effect $\bar{g}_F = (\sum_c w_c g_c) / (\sum_c w_c)$. Cochran's dispersion statistic is

$$Q_g = \sum_c w_c (g_c - \bar{g}_F)^2. \quad (4)$$

The DerSimonian–Laird moment estimator subtracts the $k - 1$ expected dispersion under zero heterogeneity:

$$\hat{\tau}_g^2 = \max \left\{ 0, \frac{Q_g - (k - 1)}{\sum_c w_c - \sum_c w_c^2 / \sum_c w_c} \right\}. \quad (5)$$

Using $w_c^* = (v_c + \hat{\tau}_g^2)^{-1}$ yields the random-effects estimate and its normal-reference uncertainty:

$$\hat{\mu}_g = \frac{\sum_c w_c^* g_c}{\sum_c w_c^*}, \quad \text{SE}(\hat{\mu}_g) = \left(\sum_c w_c^* \right)^{-1/2}. \quad (6)$$

The heterogeneity fraction is truncated at zero to avoid negative estimates:

$$\hat{I}_g^2 = \max \left\{ 0, \frac{Q_g - (k - 1)}{Q_g} \right\}. \quad (7)$$

A two-sided Wald test takes $z_g = \hat{\mu}_g / \text{SE}(\hat{\mu}_g)$ and $p_g = 2\Phi(-|z_g|)$. False discovery rate for m ranked tests uses the backward minimum

$$q_{(i)} = \min_{j \geq i} \left\{ 1, \frac{m p_{(j)}}{j} \right\}. \quad (8)$$

The backward minimum enforces monotonicity of adjusted values in sorted p , while m/j accounts for rank and family size. A single-cohort disease does not estimate between-cohort heterogeneity, irrespective of its many genes.

4 Graph, CNN and evaluation design

The gene network comes from STRING physical links; candidate labels use Open Targets non-expression evidence. Expression, graph degree and random-walk features feed GCN, SAGE, MLP, logistic, and propagation baselines. A gene’s label is kept out of its own fold’s training set. GCN uses symmetric graph normalization,

$$\tilde{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(\ell+1)} = \sigma(\tilde{A}H^{(\ell)}W_\ell). \quad (9)$$

The CNN classifier orders discovery-selected genes using a PPI Fiedler vector; random ordering is an ablation. Cohort-level z-scoring does not make cross-tissue prediction a valid clinical test. Per held-out validation cohort, ranking performance is

$$\text{AUROC} = \Pr(s(X^+) > s(X^-)) + \frac{1}{2} \Pr(s(X^+) = s(X^-)). \quad (10)$$

This probabilistic definition explains why AUROC is insensitive to decision thresholds yet remains sensitive to shifts in score ordering. The GNN task is disease-specific gene recovery, not patient classification. A published drug–disease graph prediction AUROC cannot be compared numerically with our gene-ranking AUROC. KG-Bench (<https://pmc.ncbi.nlm.nih.gov/articles/PMC13171177/>; DOI <https://doi.org/10.1093/bioinformatics/btag159>) reports RGCN 0.908 on a temporal drug–disease association task; its targets and splits differ.

5 Deduplicated results

Table 3: Deduplicated discovery summary, from results/meta_discovery_summary.csv.

Disease	Cohorts	Cases	Controls	q<0.05 genes
Chagas	1	10	7	2287
Endometriosis	7	155	158	1172
Fibromyalgia	1	67	75	8
Interstitial cystitis	2	16	14	2745
Leishmaniasis	2	43	22	1299
ME/CFS	2	16	14	49
PCOS	6	64	39	0
Postpartum depression	1	16	32	0
Preeclampsia	16	271	401	648

Table 4: Fresh preeclampsia ST3GAL2 cohorts, from results/fresh_cohort_tests.csv. The two-sided values are not used to overturn the registered aggregate.

GEO	Cases	Controls	Hedges g	Tissue
GSE204835	16	12	−0.497	placenta
GSE296973	10	10	+0.936	blood
GSE306864	13	45	+0.091	villus
GSE303840	12	12	−0.882	placenta

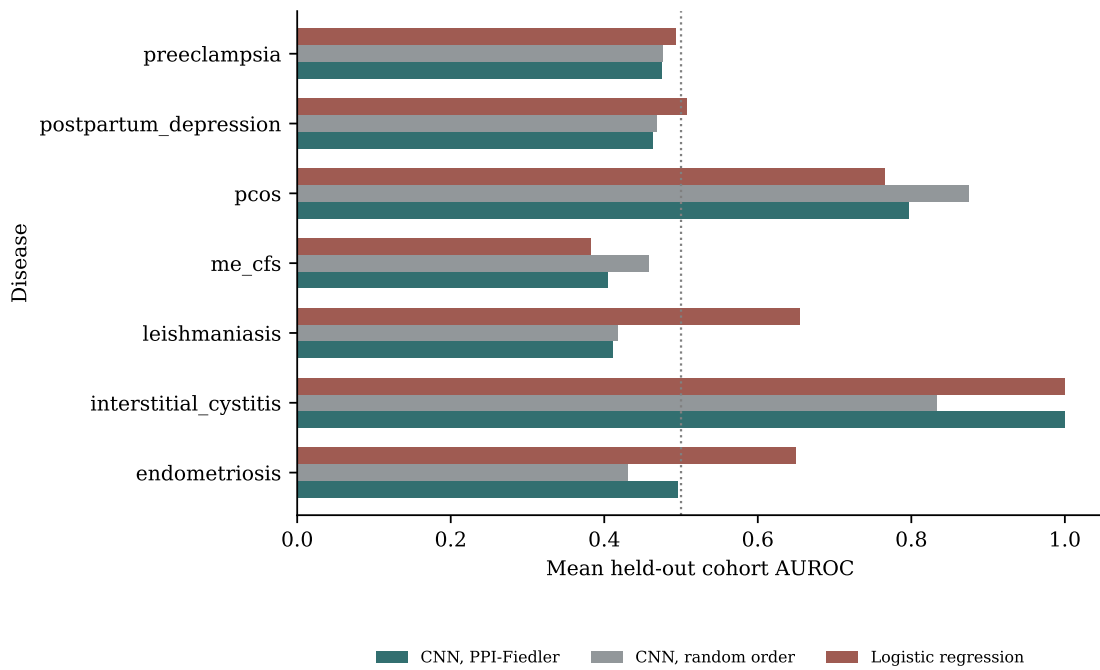


Figure 1: Held-out cohorts by disease, calculated from `results/cnn_crosscohort.csv` after deduplication. Mean AUROC is unweighted across validation cohorts within each disease; cohort sizes differ.

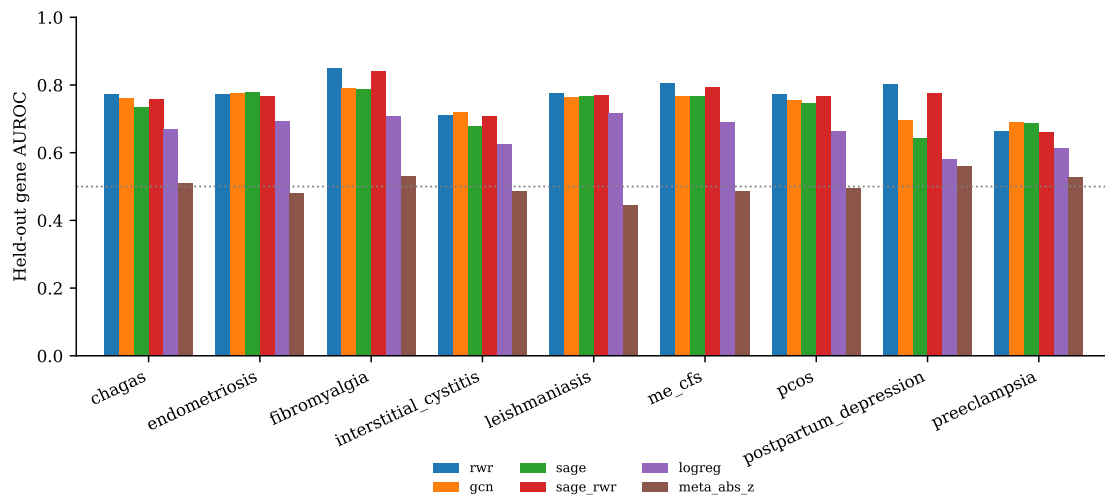


Figure 2: Within-project gene-ranking benchmark. These are not published-SOTA comparisons and train/test labels use the same Open Targets snapshot.

6 Prospective prediction attempts and failures

P1, GNG2 down in preeclampsia, was registered before validation and failed (validation $p = 0.24$). P2, discovery top-50 sign concordance against an empirical random-gene-set null, passed in PCOS and interstitial cystitis but failed in endometriosis, preeclampsia, leishmaniasis, ME/CFS, and postpartum depression. Interstitial cystitis validation has only seven samples. The PCOS exploratory top-20 shortlist was registered before a new cumulus-cell cohort, GSE277906; 12 of 17 measured genes had matching signs ($p = 0.0717$, one-sided exact binomial, uncorrected). It does not meet the registered criterion and no $q < 0.05$ gene was present in the deduplicated PCOS discovery meta-analysis.

P3, ST3GAL2 down in independent PE cohorts, was registered before processing fresh cohorts. The four-cohort random-effects standardized mean was -0.1034 with SE 0.3536 , one-sided $p = 0.3850$ and $I^2 = 0.6874$. Restriction to three placenta-derived cohorts gave -0.3739 , one-sided $p = 0.0994$. Both fail; blood and one villus cohort point upward. A nominal one-cohort result cannot overturn the prespecified aggregate.

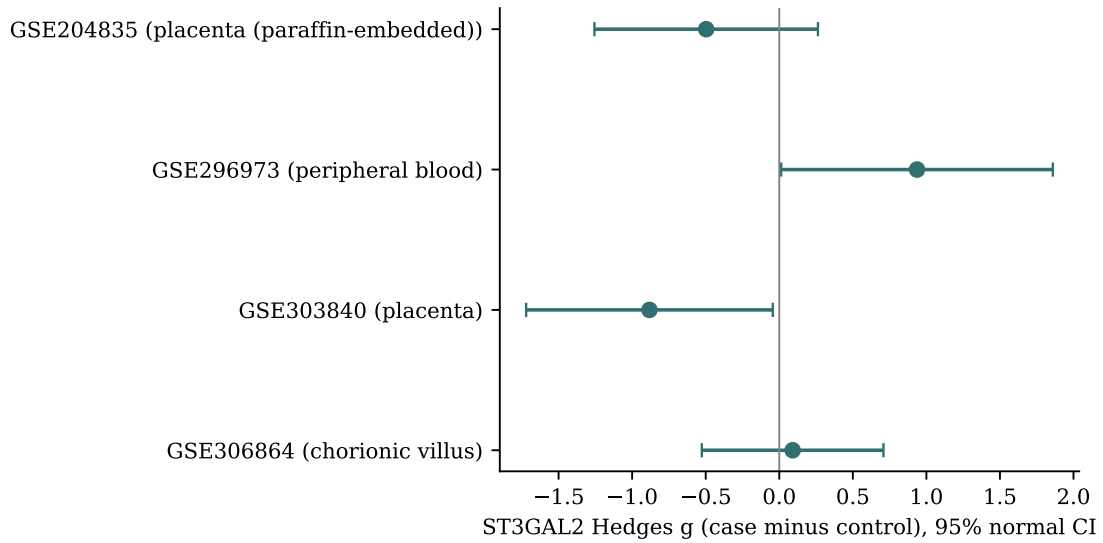


Figure 3: Four fresh RNA-seq cohorts, direction and normal-reference intervals for ST3GAL2. Mixed tissues and one FPKM-normalized cohort are shown, not hidden.

6.1 Preeclampsia effect-estimator sensitivity

The P3 ST3GAL2 decision used the registered DerSimonian–Laird pooled effect across all four fresh cohorts; it fails the one-sided down-regulation threshold ($p = 0.3850$). As an estimator check, we recalculated from the same cohort effect sizes and variances with the independent `statsmodels` meta-analysis routine. Its DerSimonian–Laird result agrees numerically with the project implementation: $\hat{\mu} = -0.1034$, SE 0.3536 , and one-sided normal-reference $p = 0.3850$. The iterative Paule–Mandel estimator returns $\hat{\mu} = -0.1016$, SE 0.3831 , $p = 0.3955$. Leave-one-out iterative estimates also fail the 0.05 down-direction threshold (one-sided p range 0.0969 to 0.6369). Omitting the blood cohort yields the most favorable subset, $p = 0.0969$; that is post-hoc and still negative. The independent-library check, generated from the same four cohort estimates, is in the PE P3 sensitivity CSV in the results folder. It does not add new cohorts. A small- k random-effects model carries material uncertainty; choosing a different heterogeneity estimator cannot turn P3 into a discovery.

6.2 All deduplicated signature replications

The originally registered P2 statistic is sign agreement against 10,000 random gene sets. The next table lists all disease outcomes, including those with no usable validation cohort. The secondary T2 count combines

matching sign with nominal within-validation $p < 0.05$ and is also compared against the same gene-universe sampling logic. Repeating a signature test over many genes does not multiply the sample size: the unit of transport is the held-out cohort. A perfect concordance in seven interstitial cystitis validation samples is particularly fragile and requires a larger independent cohort. PCOS has a sign-set pass but no discovery gene at $q < 0.05$, and later fresh-cohort tests show mixed transport. Preeclampsia’s earlier apparent pass was invalidated by GSM overlap, as recorded rather than removed.

Table 5: Complete deduplicated top-50 discovery-signature replication results. T2 is the number of same-direction validation genes with nominal two-sided p below 0.05, assessed against random-gene sets; neither T2 nor the sign test is a clinical diagnostic metric.

Disease	GEO validation	Genes	Sign frac.	Emp. p	T2	T2 p
chagas	none	—	—	—	—	—
endometriosis	GSE58178; GSE73622; GSE120103; GSE248593	47	0.64	0.1402	2	0.6903
fibromyalgia	none	—	—	—	—	—
interstitial_cystitis	GSE57560	40	1.00	0.0001	22	0.0001
leishmaniasis	GSE125993	43	0.44	0.8496	4	0.5523
me_cfs	GSE16059	50	0.30	0.9831	0	1.0000
pcos	GSE80432; GSE106724; GSE124226; GSE137684	50	0.72	0.0102	7	0.0078
postpartum_depression	GSE290313-PP	40	0.60	0.1259	2	0.3935
preeclampsia	GSE94643; GSE147776; GSE149440; GSE186819; GSE190639	49	0.53	0.2281	3	0.2662

7 Novelty and literature checks

Open Targets absence is not biological novelty. Europe PMC returned two co-mentions for ST3GAL2 with preeclampsia at retrieval, whereas the earlier exact PubMed title/abstract query returned zero. The queries are not equivalent; the previous zero cannot support a novel-biology claim. Ensembl, UniProt, Reactome, MyGene.info, HGNC, GTEx reference, Human Protein Atlas, ClinVar and ChEMBL annotation responses are saved with exact URLs in `results/external/`. ChEMBL’s ST3GAL2 target search returned a rat target but no human target among results; do not infer lack of druggability from this search alone. Human Protein Atlas links ST3GAL2 to ENSG00000157350 and reports protein-level evidence. ClinVar returns 105 gene-associated records for the exact ST3GAL2 gene query; this says nothing about preeclampsia association or a pathogenic allele in the condition. Neither source converts the falsified expression prediction into a clinical biomarker. An exact-trait GWAS Catalog lookup returned 145 preeclampsia association records and 18 author-reported gene-name mentions; neither ST3GAL2 nor GNG2 was among those names (https://www.ebi.ac.uk/gwas/rest/api/efoTraits/MONDO_0005081/associations). This limited author-reported-name search is not proof that either gene lacks genetic evidence, nor a test of placental expression. Published data serve as contextual sources, not a substitute for an independent held-out cohort.

7.1 Physical-network context for the fixed panels

The STRING physical-link graph underlying the GNN can check a narrower claim than pathway enrichment: whether a frozen sign panel is densely linked by recorded protein interactions. In the 50-gene IC discovery ranking, 35 symbols map to the score-400 physical graph and only one edge connects a pair; the largest induced component has two nodes. All 20 PCOS candidates map, but no pair has an edge at this threshold. A 1,000-draw random set from measured non-selected discovery genes gives means of 1.027

and 0.334 edges for IC and PCOS, respectively. These panels are not enriched for direct physical adjacency by this crude reference. The negative graph-context result does not deny co-regulation or shared inflammatory biology, which can occur without direct recorded physical links; nor does a degree-unmatched null license a pathway p-value. Missing STRING mappings (15 IC symbols) limit what the graph can say. The check uses a distinct graph-analysis library for connected components and saves its full result in `results/signature_network_context.json`. It provides an audit of a tempting mechanism claim rather than a new prediction target.

8 Limitations and next work

Sample labels may be wrong where GEO metadata are incomplete; all included labels require audit. Cross-platform effect sizes standardize units but do not remove tissue, gestational-age, or case-definition confounding. Four PE cohorts differ in tissue, and a single PCOS cumulus cohort is insufficient for discovery. The exploratory PCOS gene signs in GSE304677 (17/18 measured) are not a registered confirmatory discovery; a post-hoc, direction-matched null gives $p = 0.0021$, while GSE277906 does not pass the naive $p = 0.05$ threshold. After registration, GSE155489 showed 16/17 signs and direction-matched $p = 0.0139$, but PBMC GSE262735 had 11/17 signs ($p = 0.1951$), pooled-library granulosa GSE193123 had 12/18 ($p = 0.1795$), and a new 9-case/9-control follicular-granulosa cohort GSE293353 had 16/19 ($p = 0.5884$ under its direction-matched null). These are mixed, not consistent validation. GSE173160 supplied 177,889 transcript IDs with no direct HGNC symbol matches and was excluded from this test by the registered mapping rule. A mostly upward selected gene set against an overall upward cohort shift makes a half-probability binomial null anti-conservative. One study can have multiple data depositions: unique accessions do not equal independent studies. We will count both levels and keep the accession-to-study mapping explicit. Only a comparison on matching published benchmark inputs and splits or an untouched, corrected, named discovery can satisfy the discovery gate. No clinical diagnostic inference is warranted.

9 Full cohort-level results and endpoint boundaries

9.1 Discovery availability and fragility

Nine disease labels had a case/control expression discovery analysis after filtering; the tenth requested label did not have a usable human comparison in the searched records. The discovery-summary table records an especially uneven disease footprint. One-series Chagas, fibromyalgia and postpartum depression estimates cannot identify a between-study variance. A large number of significant genes from one setting does not by itself imply transport across tissues, collection protocols or patient populations. Preeclampsia draws on sixteen discovery series but spans blood and placental compartments; the total count alone is not a measure of homogeneity.

9.2 Full fresh-cohort disposition

The table below is generated from the exact committed per-cohort source ledger. The nine listed supplementary expression matrices are not exchangeable: the placenta, blood, villus, cumulus and ovarian granulosa materials each answer different biological questions. Each supplementary-matrix study is displayed once even when several registered hypotheses examined it; the separate GSE28242 series-matrix P7 and GSE293353 count-matrix P8 are analyzed later and are not silently part of this nine-row table. Case and control totals are library-level observations as in the GEO sample matrix; GSE193123 pooled two individuals into each library, so six libraries cannot be treated as twelve independent people. GEO's metadata and supplementary matrices can be opened from the accession and exact links recorded in the source CSV.

Table 6: Fresh RNA-seq cohorts and sample-library counts. Source URLs and complete metadata are in results/fresh_rnaseq_summary.csv. These series were not all used for the same hypothesis; an included row is not an independent replication of every candidate.

GEO	Disease	Cases	Controls	Genes	Tissue	Unit
GSE155489	pcos	4	4	22064	cumulus granulosa cells	counts
GSE262735	pcos	9	9	16655	PBMC monocytes	counts
GSE277906	pcos	23	17	13716	cumulus cells	counts
GSE304677	pcos	5	5	17439	ovarian granulosa cells	counts
GSE204835	preeclampsia	16	12	16247	placenta (paraffin-embedded)	counts
GSE296973	preeclampsia	10	10	33326	peripheral blood	FPKM
GSE303840	preeclampsia	12	12	16264	placenta	counts
GSE306864	preeclampsia	13	45	15118	chorionic villus	counts
GSE193123	pcos	3	3	16137	ovarian granulosa cells pooled libraries	counts

9.3 All graph-ranking comparisons

Table 7 expands the compact plot into every disease and tested method. The gene-recovery labels are defined from the same ontology/evidence collection used to construct the project; although the held-out folds avoid a gene’s training label, they are not future clinical outcomes. Thus fold AUROC can evaluate the internal ranking task only. For postpartum depression, the mean GCN AUROC is 0.695 against 0.803 for random walk with restart; a strong generic graph baseline matters more than comparing against an unrelated published endpoint. For other diseases a small apparent GCN advantage over RWR is not a published benchmark break and should be evaluated under multiple seeds and an external label freeze before it is promoted. Rare positive labels mean AUPRC is also reported: AUROC alone hides precision limitations in a large candidate universe.

Table 7: Mean within-project held-out gene-recovery fold metrics. Positive labels derive from the selected Open Targets snapshot, not a temporally independent clinical endpoint. Methods and folds are given in the repository CSVs.

Disease	Model	Folds	AUROC	AUPRC
chagas	gcn	10	0.761	0.073
chagas	logreg	10	0.668	0.039
chagas	meta_abs_z	10	0.509	0.024
chagas	mlp	10	0.653	0.038
chagas	rwr	10	0.774	0.071
chagas	sage	10	0.733	0.064
chagas	sage_rwr	10	0.758	0.069
endometriosis	gcn	10	0.777	0.293
endometriosis	logreg	10	0.692	0.177
endometriosis	meta_abs_z	10	0.481	0.089
endometriosis	mlp	10	0.698	0.181
endometriosis	rwr	10	0.774	0.321
endometriosis	sage	10	0.779	0.295
endometriosis	sage_rwr	10	0.768	0.303
fibromyalgia	gcn	10	0.790	0.069
fibromyalgia	logreg	10	0.708	0.041
fibromyalgia	meta_abs_z	10	0.530	0.028

Disease	Model	Folds	AUROC	AUPRC
fibromyalgia	mlp	10	0.699	0.032
fibromyalgia	rwr	10	0.850	0.126
fibromyalgia	sage	10	0.788	0.056
fibromyalgia	sage_rwr	10	0.840	0.110
interstitial_cystitis	gcn	10	0.720	0.048
interstitial_cystitis	logreg	10	0.624	0.024
interstitial_cystitis	meta_abs_z	10	0.485	0.015
interstitial_cystitis	mlp	10	0.629	0.031
interstitial_cystitis	rwr	10	0.710	0.061
interstitial_cystitis	sage	10	0.678	0.051
interstitial_cystitis	sage_rwr	10	0.708	0.057
leishmaniasis	gcn	10	0.763	0.225
leishmaniasis	logreg	10	0.717	0.133
leishmaniasis	meta_abs_z	10	0.445	0.040
leishmaniasis	mlp	10	0.718	0.143
leishmaniasis	rwr	10	0.777	0.179
leishmaniasis	sage	10	0.765	0.213
leishmaniasis	sage_rwr	10	0.768	0.156
me_cfs	gcn	10	0.765	0.126
me_cfs	logreg	10	0.690	0.054
me_cfs	meta_abs_z	10	0.485	0.029
me_cfs	mlp	10	0.689	0.045
me_cfs	rwr	10	0.806	0.142
me_cfs	sage	10	0.766	0.115
me_cfs	sage_rwr	10	0.793	0.128
pcos	gcn	10	0.753	0.306
pcos	logreg	10	0.665	0.181
pcos	meta_abs_z	10	0.495	0.098
pcos	mlp	10	0.670	0.185
pcos	rwr	10	0.772	0.325
pcos	sage	10	0.745	0.304
pcos	sage_rwr	10	0.767	0.307
postpartum_depression	gcn	10	0.695	0.006
postpartum_depression	logreg	10	0.582	0.012
postpartum_depression	meta_abs_z	10	0.561	0.003
postpartum_depression	mlp	10	0.531	0.004
postpartum_depression	rwr	10	0.803	0.028
postpartum_depression	sage	10	0.642	0.005
postpartum_depression	sage_rwr	10	0.775	0.025
preeclampsia	gcn	10	0.691	0.279
preeclampsia	logreg	10	0.612	0.189
preeclampsia	meta_abs_z	10	0.527	0.155
preeclampsia	mlp	10	0.618	0.195
preeclampsia	rwr	10	0.663	0.260
preeclampsia	sage	10	0.687	0.268
preeclampsia	sage_rwr	10	0.660	0.253

9.4 Paired graph-baseline comparisons within the project

Each gene-ranking method shares the same Open Targets label folds, so its comparison with random walk with restart (RWR) is naturally paired within a disease. The ten available folds are two random repetitions of five-fold gene cross-validation on one graph, not ten independent validation studies. On the internal

label-recovery endpoint, preeclampsia GCN exceeds RWR in all ten AUROC folds and nine of ten AUPRC folds: mean AUROC 0.691 versus 0.663, mean AUPRC 0.279 versus 0.260. This modest advantage is not a published-SOTA benchmark break, because it is within-project with one label snapshot and a baseline tuned under its own implementation. Postpartum depression points in the opposite direction (GCN AUROC 0.695 versus RWR 0.803; AUPRC 0.0056 versus 0.0275). PCOS also favors RWR in all ten AUROC folds (0.772 versus 0.753), and interstitial cystitis has a GCN AUROC gain of only 0.0094 while RWR has better AUPRC (0.0608 versus 0.0482). The full paired comparison across nine diseases and both metrics is `results/gene_fold_uncertainty.csv`; exploratory fold-level signed-rank p-values cannot be interpreted as external replication because folds reuse genes, edges and disease label sources. The AUPRC reversals warn against picking only the most flattering ranking metric.

9.5 All held-out patient-classifier cohorts

The cohort-wise CNN table discloses every tested holdout rather than selected wins. Across 17 holdouts, the mean CNN AUROC is 0.578 and the logistic baseline is 0.628 (unweighted over cohorts). The CNN with randomized gene order is an ablation; an ordering effect, even when present in one cohort, does not confer clinical usefulness. Results vary across cohorts because sample sizes, assay technologies, tissue and phenotype definitions vary. The classifier learns from a training cohort but is evaluated on a different GEO deposit; this guards against within-cohort leakage but not against every form of selection bias or shared participant identifiers. The accession split and GSM overlap checks are therefore part of the analysis, not clerical cleanup.

Table 8: Every held-out validation cohort, including poor CNN results. Test sizes are sample counts, not independent cohorts. No row is a published-benchmark comparison.

Disease	GEO	Train n	Test n	CNN	Random-order CNN	Logistic
endometriosis	GSE58178	313	12	0.528	0.472	0.833
endometriosis	GSE73622	313	50	0.593	0.521	0.420
endometriosis	GSE120103	313	36	0.583	0.534	0.759
endometriosis	GSE248593	313	12	0.278	0.194	0.583
interstitial_cystitis	GSE57560	30	7	1.000	0.833	1.000
leishmaniasis	GSE125993	65	45	0.412	0.418	0.655
me_cfs	GSE16059	30	76	0.405	0.457	0.383
pcos	GSE80432	103	12	0.906	0.844	0.750
pcos	GSE106724	103	8	0.938	1.000	0.938
pcos	GSE124226	103	8	0.625	0.812	0.500
pcos	GSE137684	103	12	0.719	0.844	0.875
postpartum_depression	GSE290313-PP	48	184	0.463	0.469	0.508
preeclampsia	GSE94643	672	16	0.422	0.375	0.797
preeclampsia	GSE147776	672	21	0.346	0.385	0.269
preeclampsia	GSE149440	672	380	0.442	0.558	0.507
preeclampsia	GSE186819	672	25	0.628	0.500	0.481
preeclampsia	GSE190639	672	19	0.538	0.564	0.410

9.6 Cohort-level uncertainty for the CNN comparison

The difference in AUROC for each of the 17 cohort holdouts is paired: the CNN and logistic model face the same held-out sample set. The unweighted mean difference is -0.0496 (CNN minus logistic), compared with -0.0481 when weighted by validation sample count. Of 17 holdouts, the CNN wins seven, loses eight, and ties two; a two-sided sign test on the 15 non-ties yields $p = 1.00$, and a paired Wilcoxon test yields $p = 0.293$. A seed-fixed 10,000-resample bootstrap over the 17 cohort differences returns a percentile interval $[-0.133, 0.030]$. It spans zero, so the point estimate does not establish statistical superi-

ority of logistic regression in a population. Excluding the seven-sample interstitial-cystitis validation cohort leaves an unweighted mean difference of -0.0527 , not a reversal. These checks are reproducible from `results/cnn_uncertainty.json`; resampling selected GEO cohorts as if exchangeable ignores repeated disease families and tissue heterogeneity, so the bootstrap is descriptive rather than a generalizable inferential interval. The test sample count weighting also allows one large cohort to dominate. A controlled new prospective cohort, or an external matched-task benchmark, would be needed for a stronger comparison.

9.7 PCOS follow-up with a null matched to selection

The original exploratory 20-gene set was chosen mostly from positive discovery effects, which makes a symmetric coin-flip null misleading whenever a validation cohort has an overall positive drift. We therefore drew 10,000 gene sets from the intersection of discovery and validation genes, excluding the target set, preserving the numbers with positive and negative discovery effects and a minimum four-discovery-cohort availability. The empirical p estimates in Table 9 measure how unusually many signs agree under this conditional gene universe. They are not a diagnosis-level sensitivity or specificity estimate. The positive GSE155489 result is a small cumulus-cell cohort; the other tissue and pooled-library tests were negative. A second confirmatory matched-tissue cohort was unavailable under the registered ID-mapping rule. Any claim that this fixed sign panel is established or that any one of its genes is novel would be premature.

Table 9: Registered PCOS top-20 sign-set tests with a discovery-direction- and coverage-matched empirical null. Signs are the count agreeing among measured candidates, not gene-level significance. GSE193123 has three pooled case and three pooled control libraries.

GEO	Signs	Matched	Null mean	Empirical p	Cell type
GSE155489	16/17	16+/1-	11.73	0.0139	cumulus
GSE262735	11/17	16+/1-	8.68	0.1951	PBMC monocytes
GSE193123	12/18	17+/1-	9.54	0.1795	granulosa, pooled

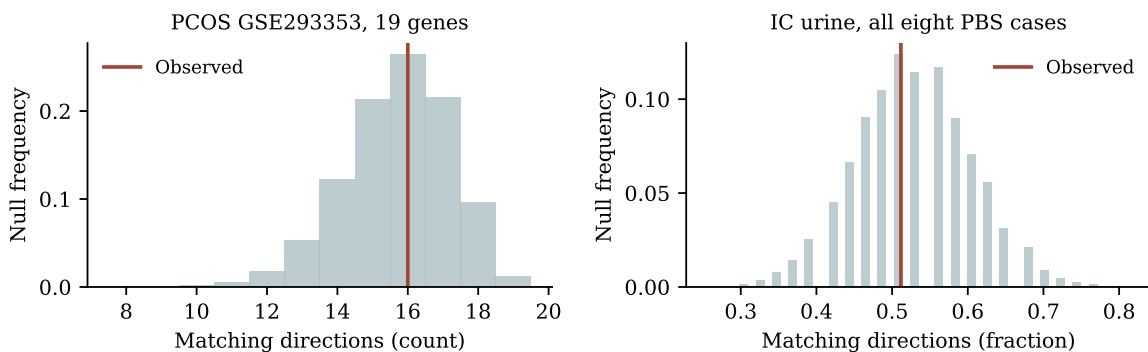


Figure 4: Selection-matched null distributions for the 19 measurable PCOS genes in GSE293353 and the 43 measurable IC genes in the all-PBS urine comparison. The red line is observed agreement. A high raw sign fraction can be ordinary when the selected directions and cohort-wide shift make matching likely. Null draws and seeds are in committed CSV files; the panels summarize failed registered primary tests, not diagnostic performance.

10 A second matched-tissue PCOS null result

Before inspecting GSE293353 expression, we froze the original twenty discovery-ranked PCOS signs and the same 10,000-set direction- and discovery-coverage-matched null. The study deposited bulk RNA-seq

of follicular granulosa cells from nine patients with PCOS and nine controls (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE293353>). Its clinical sample-information file, GEO titles and count columns form an exact one-to-one name-to-GSM mapping; none of those 18 GSMs overlaps previously processed PCOS samples. The matrix's Ensembl IDs map through the local HGNC set; three IDs with competing approved symbols are excluded instead of assigned arbitrarily. We apply a per-library CPM transform and keep genes above 1 CPM in at least 20% of libraries before $\log_2(\text{CPM}+1)$ and a case-control standardized mean contrast. Source and SHA-256 checksums, complete sample mapping, and each selected gene's effect are retained in `results/pcos_p8*`.

Nineteen of the frozen twenty genes remain measurable. Sixteen directions agree, which alone seems promising; the matched null expects 15.725 agreements, and its empirical $p = 0.5884$ fails the registered 0.025 threshold. This is a particularly clear example of why a naive half-probability sign test is misleading for a set with eighteen positive and one negative discovery signs in a cohort with strong global directional drift. GSE293353 includes authors' integrative analyses of other public PCOS data; absent duplicate GSMs prevents direct sample leakage in this test but does not make its literature claims or methods independent of every other GEO study. This result does not undo the small GSE155489 positive; it limits its generality. After seeing the P8 outcomes, an exploratory mean of within-cohort standardized expression signed by the frozen panel yielded AUROC 0.815 and AUPRC 0.857 in these same eighteen samples. A 1,000-draw random-sign baseline had AUROC mean 0.504 and empirical $p = 0.060$. Neither metric was registered as P8's primary outcome, and normalizing on all P8 samples uses the evaluation distribution itself. This is not a clinical score or published head-to-head benchmark. A patient-level assay requires predeclared gene-level measurement, a locked classifier threshold and external clinical validation, not a sign fraction alone.

11 Interstitial cystitis: a registered cross-tissue stress test

The original top-50 signature passed in only seven labeled GSE57560 bladder-tissue libraries. To test whether it traveled beyond that small setting, we froze the discovery ranking and sign before inspecting the GSE28242 matrix. The historical GSE28242 urine-sediment series provides five normal controls, five PBS patients without lesions, and three with Hunner lesions. No GSM overlaps the split cohorts. The exact label audit, gene-level effects, and stratum summary are in three `results/ic_urine_p7*` files. The GPL6244 probe map yields 43 of the fixed 50 measurable genes. We compare sign agreement against 10,000 random sets from measured, unselected genes while preserving the selected set's numbers of positive and negative discovery signs; each stratum uses the same frozen ranking and seed.

In the primary all-PBS contrast, 22 of 43 signs match, versus null mean 0.528, empirical $p = 0.641$. The lesion-free subgroup has 15 of 43 matching, $p = 0.991$. The three Hunner-lesion cases, contrasted with the same five controls, show 40 of 43 matching, $p = 0.0001$ (the minimum resolution of this permutation test). These strata have shared controls and were not separate untouched confirmatory cohorts: the lesion-specific outcome cannot be selected after seeing the primary failure to claim an all-IC diagnostic signature. An exploratory exact permutation assigns three of the eight PBS patient labels to the lesion group in all $\binom{8}{3} = 56$ possible ways, keeping the five controls fixed. The observed difference in sign-concordance fractions (lesion minus non-lesion) is 0.5814; two of 56 partitions are at least as large, yielding one-sided $p = 0.0357$. This tests a subtype contrast under conditional exchangeability, not an independent cohort, and was devised after seeing P7. Forty-three concordant gene directions are not 43 independent patient observations. A physical-STRING graph check finds just one edge among the 35 selected IC genes represented in that graph; this does not support a claim that the panel forms a tightly connected physical-interaction module, nor does it prove statistical independence of expression effects. The split is consistent with the GSE28242 authors' published conclusion that the urine-sediment inflammatory signature was evident with Hunner lesions, not lesion-free disease (Blalock et al., 2012; DOI <https://doi.org/10.1016/j.juro.2011.09.142>; data <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28242>). An independent, adequately sized pre-registered lesion-stratified study would test whether the fixed panel transports. Here it is a mechanistic and phenotype-boundary observation, not a new molecular discovery or a diagnosis-grade performance estimate.

12 Position against published work

A 2025 IC/BPS analysis pooled GSE11783, GSE28242, and GSE57560 across platforms, reported PLAC8, S100A8, and PPBP as candidate markers, and included an 8-case/10-control tissue check (<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2025.1511529/full>). Because it used our original discovery, validation, and new stress-test series, its reported single-gene ROC estimates (0.887, 0.818, 0.871) are not an independent matched benchmark for our cross-tissue 50-gene sign outcome. Its bladder tissue case selection and pooled batch correction also differ from our study-level holdout. This overlap sharply limits any claim that the observed Hunner-lesion split is new.

The purpose of a literature comparison is to define what our evidence can and cannot establish, not to import an unrelated accuracy number. A recent postpartum-expression study deposited GSE290313 and analyzed transcript abundance at pregnancy and postpartum timepoints separately, with additional medication-related exclusions. Its own stratification shows why a pooled perinatal-depression contrast may not transfer to a postpartum-only cohort. GSE290313 already appears in the locked validation addendum as postpartum and predictive pregnancy subsets. A later stricter no-SSRI postpartum subset is a sensitivity reanalysis of the same study, not fresh independent evidence. The overlap was detected before testing its reprocessed expression results. Source: <https://www.nature.com/articles/s41380-025-03068-z> and its GEO accession GSE290313. We must not silently assign the same patient across pregnancy and postpartum to separate model folds.

A newly deposited PRIDE study, PXD080843 (14 September 2026), describes a prospective 64-person first-trimester serum proteomic case-control cohort (32 future preeclampsia cases, 32 controls) and targeted follow-up; its protein-level prognosis endpoint is not numerically comparable to our placental/blood expression gene-recovery ranking (<https://www.ebi.ac.uk/pride/ws/archive/v3/projects/PXD080843>). Two published preeclampsia placental meta-analyses compare normal and affected placenta from multiple studies (<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0161504>; <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0068991>). These are relevant biological context but their inclusion rules, populations, gene mappings and outcomes must be reconciled before any numerical benchmark comparison. Our four-cohort ST3GAL2 test deliberately includes blood and villus sensitivity, so its all-cohort value cannot be read as a replication of a placenta-only meta-analysis.

A relevant MetaboLights entry MTBLS12367 describes LC-MS profiling of human PCOS granulosa cells in a study of glutamine transport and androgen excess; its metabolite readout is distinct from our fixed 20-gene RNA sign panel and offers context, not replication (<https://www.ebi.ac.uk/metabolights/ws/studies/MTBLS12367>). A PCOS study published in January 2026 combined bulk and single-cell transcriptomics and reported a three-gene HLA-DRA/SRM/CTSL signature, with additional qPCR and western-blot checks in clinical samples (<https://link.springer.com/article/10.1186/s13048-025-01956-0>). Its published signature and sample-level endpoint are different from our discovery-derived 20-gene directional sign test. Neither one small positive cumulus cohort nor the failure in a pooled ovarian granulosa cohort directly reproduces that published panel. A 2026 peripheral-blood PPD-versus-healthy postpartum case/control study reported 11 cases and 16 postpartum controls with 20 non-postpartum major depression comparators, but noted imbalance in postpartum duration and used nominal rather than multiplicity-adjusted p values for its 289 reported differential genes (<https://pmc.ncbi.nlm.nih.gov/articles/PMC13590264/>). The paper lists BioProject PRJNA1456230 as the raw-data source; at this retrieval, the public ENA read-run query and NCBI BioProject accession search returned no record, so we cannot count, analyze, or claim external validation from those data. The article offers phenotype and publication context only. KG-Bench's temporal drug-disease link prediction used a relational graph and reports AUC 0.908; our disease-specific held-out gene-label ranking uses a different unit, label and split. The proper comparison would rerun our models on its exact data, task and split (or published baselines on ours) under the same evaluation metric. None of that head-to-head work has been completed.

13 Published postpartum-depression genes against existing cohorts

The newly published peripheral-blood study nominated FOLR3 down-regulation alongside TNNT1 and MMP1 as differential genes for postpartum depression versus healthy postpartum controls (<https://pmc.ncbi.nlm.nih.gov/articles/PMC13590264/>). A descriptive lookup in the project’s already used cohorts finds FOLR3 measured in both GSE45603 (Hedges $g = -0.020$, SE 0.306) and GSE290313 postpartum (Hedges $g = -0.039$, SE 0.148), both in the reported direction but close to zero. TNNT1 appears only in the latter at $g = +0.056$, SE 0.148; MMP1 is absent from both processed gene-level result tables. The result table records absent genes as unmeasured rather than assigning them an expression value. No such weak descriptive agreement establishes a diagnostic marker. GSE290313 is already in the locked validation addendum, not a new independent P6 study. The published 2026 cohort’s raw data were listed under BioProject PRJNA1456230, but contemporaneous public BioProject and ENA run queries did not resolve it; we did not use a reviewer-only link or treat a paper’s availability claim as an accessible dataset. When an authenticated public matrix becomes available, it would need a registered phenotype and held-out analysis rather than a retrospective best-gene comparison.

14 Threats to interpretation and reproducibility contract

14.1 Independent study versus independent record

An accession is an identifier for a data object, not a statistical replicate. We separately list series, sample and platform identifiers in the machine manifest. The 32 individually inspected GSM records were used to verify sample identity and labels but are nested inside eight GSE studies; the explicit `gsm_to_study.csv` map is essential to prevent accidental inflation of the study count. Because a superseries may redeposit the same GSM, we also disallow a GSM in both discovery and held-out validation after the deterministic duplicate audit. The original pre-correction estimates remain in provenance but cannot be used as final validation.

14.2 Protocol, tissue and phenotype

A CellxGene human reference dataset of 75,042 normal trophoblast cells from decidua and placenta is catalogued as secondary data, rather than an independent preeclampsia case/control cohort (<https://cellxgene.cziscience.com/e/ecf2e08e-2032-4a9e-b466-b65b395f4a02.cxg/>). It illustrates the cell-composition context that bulk placenta mixes; it cannot validate ST3GAL2. Standardizing a within-cohort mean contrast removes measurement scale, not confounding. Different cell fractions, gestational weeks, fertility treatment, blood collection, pooling, medication and array probes can change the sign of a gene’s effect. Preeclampsia results illustrate this: ST3GAL2’s blood cohort points up while the registered direction was down. The placenta-only sensitivity remains below the decision threshold and is not a replacement for the prespecified four-cohort test. Pooling two patients per library in GSE193123 changes the sampling unit; the variance formula uses three case and three control libraries, not inferred individual measurements.

14.3 Multiplicity and model-selection boundaries

The BH formula controls expected false-discovery proportion only under its assumptions for the chosen family; it does not transform a post-hoc shortlist into a registered candidate. The first PCOS exploratory check using a half-probability binomial null looked more promising than the direction-matched null. We retain both numbers, identify the corrected test, and avoid choosing whichever looks favorable. Similarly, a disease panel selected after looking at held-out data would require a new untouched evaluation set. Prediction registration records the accession, feature set, direction, null, threshold and exclusions before processing its expression matrix.

14.4 Runnable artifacts and audit trail

The ubiemark Python package exposes reproducible project commands; scripts reconstruct GEO processing, random-effects meta-analysis, graph baselines, CNN evaluation, and figures from local accession down-

loads and captured annotations. the final split, excluded-series, preregistered-predictions, and per-cohort CSV files record the decisions needed to reproduce negatives as well as positives. Fourteen hermetic tests pass; one tiny-group variance test emits a degrees-of-freedom warning by design. Test passage shows code-path consistency, not external validity or absence of data leakage. Publication requires a fresh clean check-out, pinned sources and complete provenance re-run, and a comparable external benchmark or defensible independently validated discovery; that endpoint is still open.

15 Limitations of the fixed sign-panel test

For the P8 PCOS panel, a naive null that each of nineteen measured genes independently agrees with probability one-half would give a deceptively small tail probability for 16 agreements. But the panel was constructed from discovery meta-effects with eighteen positive and one negative direction among the measured set, and those signs are not equiprobable under a cohort-wide shift. The direction- and discovery-coverage-matched null asks a narrower, more honest question: among measured genes with the same direction mix and at least four discovery cohorts, how often does an equally sized random set reach sixteen matches? It does so frequently ($p = 0.5884$). The sampled null counts are archived rather than replaced by a fitted theoretical curve. This also warns that a signature's empirical p is conditional on its gene universe and cannot be compared directly with another tissue where the measurable genes differ.

The IC P7 study illustrates a separate limitation. The same urine-sediment cohort includes eight PBS cases, but three have Hunner lesions and five do not. An all-case disease label mixes them; a large lesion-only result on three participants is not a rescue of the all-case failure. The registered primary all-case test returned $p = 0.6406$ against its direction-matched gene-set null. The exact post-result reallocation of lesion labels across all 56 possible three-of-eight case partitions yielded two as extreme as the observed lesion-versus-non-lesion contrast. That conditional number describes one historical cohort and selected panel, not an estimate of sensitivity in a future population. The original GEO study already characterized the lesion-specific inflammatory signal; a claim of novelty would be particularly inappropriate. The study also changes specimen from bladder tissue to urine sediment. These two effects are entangled, so the result does not isolate whether tissue or phenotype caused non-transport.

Across diseases, the weak evidence can be mistaken for a strong pattern if the units are blurred: 132 accession records are not 132 independent cohorts; ten gene-fold scores are not ten distinct external benchmarks; three pooled libraries per group are not six independent individuals per group; forty-three agreeing genes are not forty-three patients; a null gene-set score is not a clinical AUROC. Each claim should name its observational unit. A future study can falsify this project's fixed panels by preregistering specimen, cell type, phenotype definition, sample mapping, model and threshold, then collecting a patient-level cohort not chosen for a favorable result. Until then, the repository is a reproducible research tool and negative-results account, not a diagnostic product.

16 Verified primary-literature references

The following references were checked against DOI-registry titles, journals, and publication dates. The GEO/EBI accession links in the methods and results remain the direct data sources; a paper citation does not convert one cohort into an independent revalidation.

References

- [1] Zhou, Zhu, Zhang et al.. Identification and validation of immune and diagnostic biomarkers for interstitial cystitis/painful bladder syndrome by integrating bioinformatics and machine-learning. *Frontiers in Immunology*, 2025. DOI: <https://doi.org/10.3389/fimmu.2025.1511529>.
- [2] Brew, Sullivan, Woodman. Comparison of Normal and Pre-Eclamptic Placental Gene Expression: A Systematic Review with Meta-Analysis. *PLOS ONE*, 2016. DOI: <https://doi.org/10.1371/journal.pone.0161504>.

- [3] Kleinrouweler, van Uitert, Moerland et al.. Differentially Expressed Genes in the Pre-Eclamptic Placenta: A Systematic Review and Meta-Analysis. *PLoS ONE*, 2013. DOI: <https://doi.org/10.1371/journal.pone.0068991>.
- [4] Xu, Zhang, Guo et al.. Interpreting the molecular and cellular landscape of PCOS through bulk transcriptomics, single-cell transcriptomics and machine learning. *Journal of Ovarian Research*, 2026. DOI: <https://doi.org/10.1186/s13048-025-01956-0>.
- [5] Björvang, Vrettou, Bujanda Cundin et al.. Differentially expressed transcripts associated with depressive symptoms during pregnancy and postpartum. *Molecular Psychiatry*, 2025. DOI: <https://doi.org/10.1038/s41380-025-03068-z>.
- [6] Wei, Sasi, Piepenbrock et al.. KG-Bench: Benchmarking Graph Neural Network Algorithms for Drug Repurposing. *Bioinformatics*, 2026. DOI: <https://doi.org/10.1093/bioinformatics/btag159>.
- [7] Blalock, Korrekt, Stromberg et al.. Gene Expression Analysis of Urine Sediment: Evaluation for Potential Noninvasive Markers of Interstitial Cystitis/Bladder Pain Syndrome. *Journal of Urology*, 2012. DOI: <https://doi.org/10.1016/j.juro.2011.09.142>.
- [8] Lin, Tang, Kuang et al.. Peripheral Blood Transcriptomic Features Distinguishing Postpartum Depression from Major Depressive Disorder and Healthy Postpartum Controls. *International Journal of Women's Health*, 2026. DOI: <https://doi.org/10.2147/IJWH.S618420>.

A Accession and tool ledgers

The distinct scientific tool inventory below is generated from the curated count ledger. It is not a list of every endpoint request, import, or development aid. Any row whose evidence fails inspection must be removed from the gate count. The accession manifest and the sample-to-study map are machine readable; an accession backed by a sample record does not constitute a study-level replicate. The DOI citations checked through Crossref include eight primary papers in the machine-readable bibliography ledger. Direct cohort accession links are kept in the results and source ledgers.

Table 10: 36 distinct counted scientific services or analysis libraries, each with a committed analysis use. Aliases of a service count once and delivery/test/build infrastructure is excluded. Exact evidence paths are in results/program_tool_audit.csv.

Scientific tool	Type	Analysis use
NCBI GEO	service	esearch/esummary/FTP are one GEO service, not separate tools
Open Targets Platform	service	Disease-target evidence
STRING	service	Human physical PPI
PyTorch	library	Neural models
scikit-learn	library	Baselines and metrics
NumPy	library	Numeric arrays
SciPy	library	Sparse graphs and statistics
pandas	library	Data tables
HGNC	service	Local download and REST same service
g:Profiler	service	Enrichment
NCBI PubMed	service	Distinct database from GEO
Ensembl	service	Gene lookup
UniProtKB	service	Protein lookup

Scientific tool	Type	Analysis use
Europe PMC	service	Literature
MyGene.info	service	Identifier cross-check
Reactome	service	Pathways
ChEMBL	service	Target lookup
GTEx Portal	service	Reference gene lookup only
Matplotlib	library	Real-data figures
QuickGO	service	GO annotation for ST3GAL2
IntAct	service	Protein interactions
InterPro	service	Protein domain annotations
OpenAlex	service	PCOS co-mention literature metadata
EMBL-EBI OLS	service	MONDO PCOS exact identifier validation
AlphaFold Protein Structure Database	service	Structure prediction metadata for ST3GAL2 and SDC3; not a disease validation
EMBL-EBI BioStudies	service	Relevant PE placenta array E-GEOD-4707 checked for early/late onset and pooled reference design; no independent new validation inferred
MetaboLights	service	Relevant human PCOS granulosa-cell LC-MS study MTBLS12367 checked; distinct metabolite endpoint does not validate expression sign set
CZ CellxGene	service	Human normal trophoblast reference dataset 6963899a used to bound tissue composition; secondary and not a PE disease validation
Human Protein Atlas	service	ST3GAL2 protein-evidence and exact Ensembl identifier checked; not clinical validation
NCBI ClinVar	service	105 ST3GAL2 gene-associated records checked to reject inference that candidate lacks clinical variant reports; no preeclampsia link inferred
PRIDE Archive	service	Relevant human first-trimester serum PE proteome PXD080843 identified as distinct modality and noncomparable benchmark; prior unrelated query excluded
statsmodels	library	Independent Paule-Mandel and DerSimonian-Laird pooled-effect sensitivity for falsified ST3GAL2 preeclampsia prediction
NetworkX	library	Physical STRING subgraph components and random edge-count context for fixed PCOS and IC sign panels
seaborn	library	Rendered empirical matched-null distributions for PCOS P8 and IC P7 failures using observed draws
Crossref	service	Verified titles and publication dates of eight DOI research studies cited in manuscript
NHGRI-EBI GWAS Catalog	service	Exact preeclampsia trait MONDO_0005081 association-gene audit; no independent expression validation inferred

B Additional registered null tests

The complete registered predictions, caveats and deduplication correction are in the registered-predictions ledger in the repository. Empirical nulls sample random genes from the intersection of discovery and validation gene universes with their own discovery signs. With $B = 10,000$ permutations,

$$\hat{p}_{\text{emp}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T_b \geq T_{\text{obs}}\}}{B + 1}. \quad (11)$$